

Topics to discuss

- Boole's Rule
- Numerical Problem
- Error in Integration
- Homework Problem

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Boole's Rule :

It is a higher order Newton Cotes formula.

If we take $n=4$ in Newton Cotes formula then we will get Boole's Rule.

→ Number of interval, n , must be multiple of 4

→ It is a five point formula.

The formula is given by,

when $n=4$,

$$\int_{x_0}^{x_n} f(x) dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4)$$

Where, $x_n = x_0 + 4h$

When, $n = 8$ then,

$$\int_{x_0}^{x_n} f(x) dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4 + 7y_4 + 32y_5 + 12y_6 + 32y_7 + 7y_8)$$

Where, $x_n = x_0 + 8h$

$$\& \quad h = \frac{x_n - x_0}{n}$$

Numerical Problem :

Q: Use Boole's formula to compute,

$$\int_0^{\pi/2} \sqrt{\sin x} \, dx$$

Solution: Considered, $n=4$

$$\text{Let, } f(x) = y = \sqrt{\sin x}$$

$$\therefore \text{step length, } h = \frac{b-a}{n} = \frac{\pi/2 - 0}{4} = \frac{\pi}{8}$$

$$\text{So, } x_0 = 0$$

$$x_1 = x_0 + h = 0 + \frac{\pi}{8} = \frac{\pi}{8}$$

$$x_2 = x_0 + 2h = 2 \frac{\pi}{8} = \frac{\pi}{4}$$

$$x_3 = x_0 + 3h = \frac{3\pi}{8}$$

$$x_4 = \frac{\pi}{2}$$

Lets make the table for $y=f(x)=\sqrt{\sin x}$

x	0	$\frac{\pi}{8}$	$\frac{\pi}{4}$	$\frac{3\pi}{8}$	$\frac{\pi}{2}$
$y=f(x)$	0	0.61861	0.840896	0.961186	1.00

Now, let apply the boole's rule for $n=4$

$$\int_{x_0}^{x_n} f(x) dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4)$$

$$= \frac{2\pi}{8 \times 45} (7 \times 0 + 32 \times 0.61861 + 12 \times 0.840896 + 32 \times 0.961186 + 7 \times 1)$$

$$= \frac{\pi}{180} [32(0.61861 + 0.961186) + (12 \times 0.840896) + 7]$$

$$= 1.1806175 \text{ Ans.}$$

Homework Problem:

Q: Evaluate approximately;

$$\int_0^4 \frac{1}{1+x^2} dx$$

using Boole's Rule

Solution: Let $y = f(x) = \frac{1}{1+x^2}$

considered, $n=4$

$$\text{So, Step length, } h = \frac{b-a}{n} = \frac{4-0}{4} = 1$$

$$\therefore x_0 = 0$$

$$x_1 = x_0 + h = 0 + 1 = 1$$

$$x_2 = x_0 + 2h = 2$$

$$x_3 = x_0 + 3h = 3$$

$$x_4 = 4$$

Now let's make table, $y = f(x) = \frac{1}{1+x^2}$

x	0	1	2	3	4
y	1	$\frac{1}{2}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{17}$

Now applying boole's formula for $n=4$,

$$\int_{x_0}^{x_n} f(x) dx = \frac{2h}{45} (7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4)$$

$$= \frac{2}{45} \left(7 \times 1 + \frac{32}{2} + \frac{12}{5} + \frac{32}{10} + \frac{7}{17} \right)$$

$$= 0.0445 (7 + 16 + 2.4 + 3.2 + 0.41176)$$

$$= 1.28942$$

Error :

Exact value ,

$$\begin{aligned}\int_0^4 \frac{1}{1+x^2} dx &= \left[\tan^{-1}(x) \right]_0^4 \\ &= \tan^{-1}(4) - \tan^{-1}(0) \\ &= 1.3258\end{aligned}$$

and, Approximate value, = 1.28942

$$\begin{aligned}\therefore \text{Absolute Error,} &= |\text{Exact value} - \text{Approx. value}| \\ &= |1.3258 - 1.28942| \\ &= 0.03641\end{aligned}$$

$$\text{and Relative Error,} = \frac{\text{Absolute Error}}{\text{Exact value}} = \frac{0.03641}{1.3258} = 0.02746$$

$$\begin{aligned}\therefore \text{Percentage Error,} &= (\text{Relative Error} \times 100)\% \\ &= (0.02746 \times 100)\% \\ &= 2.746 \%\end{aligned}$$

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